LESSON 9.4

Apply Two-Variable Relationships to Solve Problems

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation to represent the relationship between two variable quantities and use it to solve a real problem.

WHY THIS MATTERS An automobile plant's assembly line runs in 8-hour shifts and produces 2.5 cars per hour. An order arrives: 1,000 cars in 3 weeks. Should the manufacturer accept? Later, the student council plans a fundraiser car wash — last year they washed 34 cars and raised $408, and this year the goal is $500.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
solution *Spanish: solución*	A value of the variable that makes an equation true.	the number you solve for
substitute *Spanish: sustituir*	To replace a variable with a number so you can compute.	put 8 in for x
at least *Spanish: al menos*	That amount or more — the smallest amount that still works.	“At least $500” means $500 counts, and so does anything more.
predict *Spanish: predecir*	To make a reasonable guess about a future amount using information you already have.	We measured up to 5 minutes → 6,000 feet. Reach PAST what was measured: 6 minutes should be about 7,200.
justify *Spanish: justificar*	To explain WHY an answer or recommendation makes sense, using the mathematics.	“35 cars at $14.30 raises $500.50, which clears the $500 goal.”
Equation *Spanish: Ecuación*	A math sentence with an equals sign showing two amounts are the same.	c = 4n

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bakery makes 24 rolls each hour: r = 24h. Is 5 hours enough for 150 rolls?

1. Substitute h = 5: r = 24 × 5 = 120 rolls.
2. Compare: 120 < 150.

Answer: No, 5 hours makes only 120 rolls

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Last year the council washed 34 cars and raised $408. What did they charge per car? Divide: 408 ÷ 34 = $12.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I let _____ stand for _____ and _____ stand for _____ .
- My equation is _____ , so I substitute _____ for _____ .
- The solution is _____ , which means _____ in this situation.
- This answer is reasonable because _____ .

Word bank equation · substitute · solution · rate · variable · multiply · per · total · reasonable · check · input · output

WATCH OUT!

A common mistake

Adding the rate instead of multiplying

LESSON 9.4

Apply Two-Variable Relationships to Solve Problems

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

The assembly line produces 2.5 cars per hour. How many cars does it produce in one 8-hour shift?

(A) 10.5 cars (B) 16 cars

(C) 20 cars (D) 25 cars

STEPS TO TRY
1. How many cars come off the line in ONE hour?
2. Each of the 8 hours adds another 2.5 cars.
3. Multiply 2.5 by 8.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

2 **Circle the letter of the best answer. Show how you know.**

Let h = hours of operation and c = cars produced. Which equation represents the assembly line?

(A) c = 2.5h

(B) c = 2.5 + h

(C) h = 2.5c

(D) c = h ÷ 2.5

STEPS TO TRY

1. Say it in words: the cars produced equal 2.5 ___ the hours.
2. Test your choice against the table row (8, 20).
3. Which equation gives 20 when h = 8?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Using the table (1 hour, 2.5 cars; 8 hours, 20 cars; 16 hours, 40 cars), how many cars are produced in 24 hours?

(A) 26.5 cars

(B) 40 cars

(C) 50 cars

(D) 60 cars

STEPS TO TRY

1. How many 8-hour shifts fit in 24 hours?
2. Each shift produces 20 cars.
3. Three shifts of 20 cars each.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Last year the student council washed 34 cars and raised $408. How much did they charge per car?

(A) $10
(B) $12
(C) $34
(D) $374

STEPS TO TRY

1. The total is the price per car times the number of cars.
2. Which operation undoes that multiplication?
3. Divide 408 by 34.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

EXPLAIN YOUR THINKING

One rate is cars per hour and the other is dollars per car. Ask your partner to predict one value from each, then check by substituting — which prediction could you make without a calculator, and why?

SENTENCE STARTER Substituting ___ gives ___, so I predict ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help